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Subject: Information Security

Project Proposal

Title: SQL Injection Attacks and Their Prevention in Web Applications

1. Problem Statement / Project Idea

Web applications often store sensitive user data, but improper input validation can leave them vulnerable to **SQL Injection** attacks. These attacks allow malicious users to manipulate database queries, bypass authentication, access confidential information, and modify or delete data.

The aim of this project is to study SQL Injection attacks and implement techniques to prevent them. The project will demonstrate how vulnerabilities arise in a simple login system and how secure coding practices can mitigate these risks.

Objectives:

- Understand SQL Injection attacks and how they work.
- Demonstrate exploitation of a vulnerable login system.

- Implement prevention techniques like prepared statements, parameterized queries, and input validation.
- Highlight the importance of secure database interactions.

2. Scope

The project will focus on:

- A simple web-based login system with database interaction.
- Demonstrating SQL Injection to bypass authentication.
- Explaining basic types of SQL Injection.
- Implementing prevention mechanisms including:
    - Prepared statements
    - Parameterized queries
    - Input validation
    - Least privilege principle

3. Conclusion

SQL Injection is a serious threat to web applications. This project will demonstrate how such attacks occur and how to prevent them using secure coding practices. By comparing vulnerable and protected systems, it highlights the importance of input validation and safe database interactions. The project aims to provide practical knowledge for building secure web applications.